Rank-27 Rephase/Restart Accounting Repair after the Vivify Audit

Tactix engineering handoff

21 August 2026

Objective and outcome

The user asked that Tactix's Vivify behavior be compared with Kissat across admission, scheduling, candidate order, and interleaving with other proof rules, and requested this Andrew-facing report after the issue was fixed. The first policy defect was indeed Vivify-related: formula classification disabled Vivify for selected size ranges even though Kissat admits it unconditionally and governs work internally. That gate was removed in the preceding increment. The rank-27 investigation then continued from the latest common semantic state instead of assuming every later split was also owned by Vivify.

After the reference-wide Congruence repair, Tactix and Kissat match through the opening formula, 741,305 decisions, and conflict-2027 analysis. The first new portable divergence is a focused Restart: Kissat restarts at conflict 2027, while Tactix waits until conflict 2028. Their glue averages, admission values, deadlines, learned clause, and backtrack target are exact. The sole earlier difference is that Tactix counted Rephase's root-trail clear as a canonical Restart; Kissat does not. That one extra event crosses the shipped logarithmic interval boundary.

The smallest repair separates the shared sound backtrack mechanic from canonical Restart accounting. Rephase now uses the accounting-neutral mechanic, while scheduled Restarts retain all prior accounting and trail-reuse behavior. The repair passes its boundary unit test, restores exact trace behavior through conflict 2028, and passes the mandatory 28B+5C matched guard 66/66 with zero certificate residue. The repaired guard is materially faster in aggregate (0.7864 of the accepted pin), but the immutable rank-27 witness still times out UNKNOWN at 1,200.004 seconds. This is therefore a validated general repair, not a rank-27 closure claim.

Frozen identities and constraints

The immutable rank-27 UNSAT input is `tierD24-26/5911653e4edadec391654c8f06353f9a-oski15a01b06s_opt.cnf.xz`, compressed SHA-256 `5435b1b2...bb4e9`. Frozen Kissat solves it in 278.187 seconds. That performance reference was not rerun; only the retained trace and a pinned-source diagnostic build were used to localize semantics.

The accepted predecessor binary is `b6c991e0...fb6f`. The repaired candidate is `8f3208b1...ec47`, built from commit `713cc26d...a924` plus source diff `4c334c02...b70`. Production logic contains no benchmark name, hash, rank, filename, deadline, conflict, selected literal, per-instance arm, or tuned target constant. The test's 90-to-91 count is derived from Kissat's shipped formula and does not occur in solver policy.

Localization

Fresh post-Congruence trace job `1637041` matched all 462,390 captured Tactix decisions against the retained Kissat trace. Jobs `1637042` and `1637043` extended the compact trajectory through conflict 2011; the first apparent tail difference was only the diagnostic stop. Diagnostic Kissat extension job `1637053`, pinned to source commit `8af8e56f...92c2`, independently confirmed exact decisions and learned hashes through decision 495,135 and conflict 2011.

Paired job `1637054` found the first real search split at decision ordinal 741,306. Both solvers entered conflict 2027 with the same conflict clause, learned binary hash `be68020008e7838b`, and backtrack target 22,853. Tactix then chose literal 117122 at level 22,854; Kissat restarted and chose literal 117137 at level 1.

Lifecycle job `1637056` showed exact state through conflict 2026. Kissat emitted `restart_begin`/`restart_end` at conflict 2027; Tactix did not, then Tactix restarted at conflict 2028. Both traces contained the same single earlier Rephase at conflict 1001. Restart-state job `1637058` then compared the fast and slow glue EMAs, margin, admission result, next deadline, and restart count. Every value except the count matched. After the common Rephase, Tactix was one higher; at the decisive boundary Kissat had 90 canonical Restarts and Tactix had 91.

Source-backed cause

Pinned Kissat `rephase.c` calls `kissat_backtrack_propagate_and_flush_trail`, increments `rephased`, and never increments `statistics.restarts`. Pinned `restart.c` increments `statistics.restarts` only inside `kissat_restart`, then performs trail reuse/backtracking and updates the focused limit. Its focused interval is

$$\texttt{restartint} + \log_{10}(\texttt{restarts} + 9) - 1,$$

with shipped `restartint=1` and integer conversion at the deadline. At 90 Restarts, $\log_{10}(99) - 1 < 1$, so the added term truncates to zero and the interval remains one conflict. At 91, $\log_{10}(100) - 1 = 1$, so the interval grows to two. Old Tactix routed Rephase's neutral trail clear through `restart_apply_to(k,0)`, crossing that exact boundary one event early.

This is a general lifecycle-accounting defect, not an empirical timing threshold. It explains both the one-conflict deadline shift and why all glue EMA state could remain exact.

Implementation decision

`src/rules/restart/restart.cpp` now exposes two layers:

- `restart_backtrack_to` clamps the requested level and performs the sound trail abandonment without Restart/reuse accounting;
- `restart_apply_to` retains canonical accounting and delegates the shared mechanic.

`src/policies/default\_policy.cpp` uses the neutral layer only for Rephase's root clear. Scheduled Restarts still consume the branching-policy pick, choose their reuse level, update target phases, increment statistics, update the focused deadline, and emit the same lifecycle events. Rephase remains an H-only policy operation.

`tests/test\_rephase.cpp` pins the first interval transition. Starting at 90 canonical Restarts and conflict 2000, a Rephase clear must preserve count 90 and next conflict 2001. One real Restart must advance to 91 and next conflict 2002. This directly catches the old conflation while remaining formula- and instance-independent.

Correctness and matched guard

The clean Release build passes 69/69 tests, including proof tests and the new boundary regression. No unit or campaign leg exposed a crash or soundness defect, so the development plan's mandatory Claude repair-delegation trigger did not fire.

Repaired diagnostic job `1637060` exactly matches Kissat through the old frontier: conflict 2026's decision and Restart; conflict 2027's decision 741,305, analysis, learned hash, Restart begin/end, and next root choice 117137; and conflict 2028's decision 764,173 plus learned hash `beabd2000ac68c29`.

Only then did matched job `1637061` run the accepted and repaired pins on all 28 Tier-B fixtures plus the inherited fixed five Tier-C mechanism fixtures. The exact contract was two CPUs, 16 GiB process memory, a 1,200-second solver timeout, no node constraint, and no array throttle.

Arm	Tier B	Tier C	checked total	aggregate time
Accepted predecessor	28/28	5/5	33/33	2201.821 s
Repaired candidate	28/28	5/5	33/33	1731.598 s

The ratio is 0.786439. Each arm returns 17 SAT and 16 UNSAT results; all 34 SAT models and 32 UNSAT proofs across the pair verify. Candidate peak RSS is 3,294,292 KiB. The aggregate reports no invalid, malformed, missing, cleanup-bad, or regressed tasks and no certificate residue. Evidence is `benchmarks/runs/20260821-0255-BC-rephase-restart-accounting/`.

Rank-27 witness

Only after the guard passed was the immutable target submitted as job `1637131`. It consumed the full 1,200.004-second solver allowance and returned UNKNOWN. The runner observed a 133,169,152-byte partial proof, deleted it and all derived scratch, and published zero residue. Scheduler peak RSS was 481,552 KiB. Evidence is `benchmarks/runs/20260821-0310-tierd-rank27-rephase-restart-accounting/`.

The cumulative accepted ranks-21–40 closures therefore remain 21, 23, 24, 25, 28, 29, 34, 38, and 39. Rank 27 is not added.

Vivify conclusion, limitations, and next action

The Vivify audit produced a real policy correction: formula classification no longer disables a reference-default proof rule. The subsequent common-state comparison also shows why convergence work cannot stop at the mechanism first suspected. On rank 27, the earliest remaining portable split after corrected Congruence occurs later in search and belongs to Rephase/Restart accounting; a Vivify-specific target arm would not have repaired its producer.

This increment establishes exactness only through conflict 2028, not for the rest of rank 27's trajectory. The next concrete action is to extend the paired semantic and lifecycle trace only beyond that transaction, locate the first later common-state divergence, and implement a further change only when the source supports a general repair. Any new candidate again owes the full Tier B plus predeclared Tier C guard before another rank-27 witness.

The durable technical report is `benchmarks/reports/2026-08-21-tierd-rank27-rephase-restart-accounti` `md`. Current handoff state is in `PROGRESS.md`; durable scheduling evidence is in `docs/scheduling-parity-ledger.md` and `docs/implementation-notes.md`.